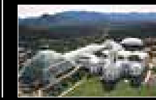

A Future Beyond Earth: Is Terraforming Our Way Out?

Soon enough the earth will no longer be able to sustain us. Do we find a new home planet or build one? That is the question.

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Misao Okawa was born in a world radically different from ours. On her island kingdom the farthest extent of technology was the occasional steamboat and mechanically configured machines; the highlight of which being firepower. Now, over 116 years later, Misao Okawa lives in Japan, the country of her birth - a country unlike the one she was born into and has witnessed change through numerous wars, natural calamities and cultural shifts.

The last hundred years have seen the Earth's population explode from 1.65 billion to over 7 billion. And this leap has come at the cost of the planet and its stability. The challenges faced by humanity, brought on by human caused climate change, such as droughts and rise in sea levels could scarcely have been imagined in the youthful life of Misao Okawa. And despite being faced with the denial of climate change by millions there is no doubt that the Earth is growing exhausted with humanity and will soon prove insufficient for our growing demands. It is perhaps time we found a new home. Or wait why not maybe create one?

Kepler-186f is the most Earth-like planet discovered but is 493 light years away, making exploration slightly problematic.

In search of Magrathea

We have marvelled over the last few months as regular updates from NASA have given us hopes for possible Earth-like planets. The Kepler mission's space telescope has searched thousands of planets and sent back observations of over 3200 candidates that could potentially be Earth-like. Of these thousands, over a 1000 planets across 400 star systems have been confirmed to be exoplanets. After hundreds of candidates over the four years of its missions scientists have a short-list of planets that could foreseeably be our next home, but in a few hundred years of course.

Most of these findings remain statistical and based on distant readings of esoteric scientific data. It is almost certain that despite the inherent importance of these discoveries they will do us little good in the short term. In the highly unlikely event that we ever did find a planet that is in a habitable orbit around a star, capable of holding liquid water and sustaining an atmosphere, we would be hard pressed to find a means to reach it. The closest Earth like planet discovered so far